

evident by now, by focusing our energies on validated learning, we can avoid much of the waste that plagues startups today. As in lean manufacturing, learning where and when to invest energy results in saving time and money.

To apply the scientific method to a startup, we need to identify which hypotheses to test. I call the riskiest elements of a startup's plan, the parts on which everything depends, leap-of-faith assumptions. The two most important assumptions are the value hypothesis and the growth hypothesis. These give rise to tuning variables that control a startup's engine of growth. Each iteration of a startup is an attempt to rev this engine to see if it will turn. Once it is running, the process repeats, shifting into higher and higher gears.

Once clear on these leap-of-faith assumptions, the first step is to enter the Build phase as quickly as possible with a minimum viable product (MVP). The MVP is that version of the product that enables a full turn of the Build-Measure-Learn loop with a minimum amount of effort and the least amount of development time. The minimum viable product lacks many features that may prove essential later on. However, in some ways, creating a MVP requires extra work: we must be able to measure its impact. For example, it is inadequate to build a prototype that is evaluated solely for internal quality by engineers and designers. We also need to get it in front of potential customers to gauge their reactions. We may even need to try selling them the prototype, as we'll soon see.

When we enter the Measure phase, the biggest challenge will be determining whether the product development efforts are leading to real progress. Remember, if we're building something that nobody wants, it doesn't much matter if we're doing it on time and on budget. The method I recommend is called innovation accounting, a quantitative approach that allows us to see whether our engine-tuning efforts are bearing fruit. It also allows us to create learning milestones, which are an alternative to traditional business and product milestones. Learning milestones are useful for entrepreneurs as a way of assessing their progress accurately and objectively: they are also invaluable to managers and investors who